

Handling Numerical Data

Height	Weight	Gender
172	150	M
180	170	M
165	140	M
190	200	M
139	100	F
145	120	F
160	140	F
172	150	F

Given $\{ H = 185, W = 170, G = ? \}$

$$P(M | H = 185, W = 170) =$$

$$P(H = 185 | M) P(W = 170 | M) P(M)$$

$$P(M) = 4/8 = 1/2$$

Assumption = H_t & W_t is Gaussian distributed random variable

Considering the 4 values of H_t & W_t from table are normally distributed and calculate μ (mean) & σ (std deviation)

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-1/2 \left[\frac{x - \mu}{\sigma} \right]^2}$$

$$P(F|H=185, W=170) =$$

$$P(H=185|F) P(W=170|F) P(F)$$

same as above

This whole process is called as Gaussian Naive Bayes when data is normally or close to normal

distribution)

IF data is not normally distributed
then use

Binomial NB

Multinomial NB

Poisson NB